CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EE) Examination

May 2013

EE402 Electrical Power System-IV

Date: 07.05.2013, Tuesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

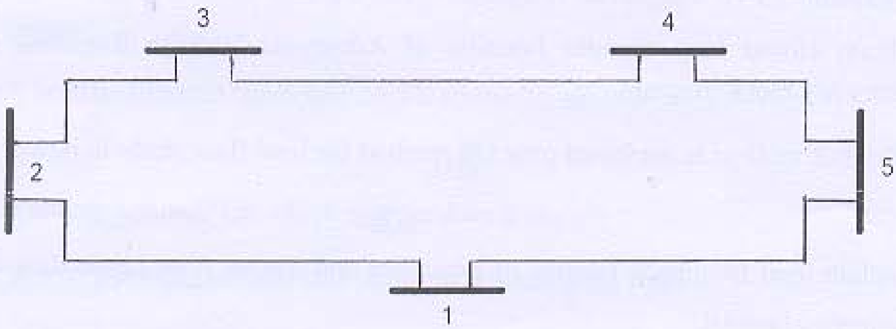
Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Give the flow chart for a load flow study on a power system having both P-Q as well as P-V buses using N-R method. **[07]**

Q - 2 (a) Each line has an impedance of 0.05+ j0.15 pu. Bus power and voltage specifications are given in the table below. **[10]**



(a) Form Y_{BUS} (b) Find Q_2, δ_2, V_3, V_4 and V_5 after 1st iteration using Gauss-Seidel method. Assume Q_2min=0.2 pu and Q_2max=0.6 pu.

Bus	P_L (pu)	Q_L (pu)	P_G (pu)	Q_G (pu)	V (pu)	Remarks
1	1	0.5	?	?	1.02/_0	Slack bus
2	0	0	2	?	1.02	PV bus
3	0.5	0.2	0	0	?	PQ bus
4	0.5	0.2	0	0	?	PQ bus
5	0.5	0.2	0	0	?	PQ bus

(b) What is economic dispatch? Explain Heat rate curve and incremental fuel cost curve of thermal power plant. **[04]**

OR

Q - 2 (a) The fuel cost of three thermal plants in S/h are given as: **[07]**

$$C_1 = 500 + 5.3P_1 + 0.004P_1^2$$
$$C_2 = 400 + 5.5P_2 + 0.006P_2^2$$
$$C_3 = 200 + 5.8P_3 + 0.009P_3^2$$

Where the powers are in MW. When the total load is 975 MW with the following generator limits (in MW):

$$200 \le P_1 \le 450$$
$$150 \le P_2 \le 350$$
$$100 \le P_3 \le 225$$

Neglecting line losses find the optimal dispatch and the total cost in \$/h by iteration. Assume the initial value of $\lambda^{(1)} = 6$.

(b) Derive the equation of transmission loss formula. **[07]**

Q - 3 (a) Obtain closed loop transfer function of Automatic Voltage Regulator loop with necessary block diagram. **[10]**

(b) Why NR method is preferred over GS method for load flow study in power system? **[04]**

OR

Q - 3 (a) Explain load frequency control of generator and derive final block diagram of load frequency control. **[10]**

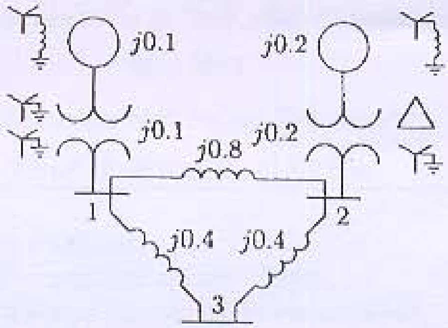
(b) Draw the final block diagram of AGC in single area system for an isolated power system. **[04]**

SECTION – II

Q - 4 Explain different types of shunt compensators. **[07]**

Q - 5 (a) A single line diagram of a simple 3 bus power system network is shown in figure. All impedances are expressed in per unit on a common 100MVA base and for simplicity resistances are neglected. Shunt capacitances are neglected and the system is considered on no load. All generators are running at their rated voltage and rated frequency with their emfs in phase. Determine the fault currents, the bus voltages and line currents during a fault using thevenin's theorem when a balanced three phase **[10]**

fault with a fault impedance $Z_f = 0.16$ per unit occurs on (a) bus 3 and (b) bus 2.



(b) Give comparison between STATCOM and SVC. [04]

OR

Q - 5 (a) Explain unbalanced fault analysis using bus impedance matrix for single line to ground, line to line and double line to ground fault. [06]

(b) What do you mean by compensation? Explain load compensation and system compensation. [08]

Q - 6 (a) Derive the expression for voltage when the line is terminated by an inductance and capacitance. [07]

(b) Explain briefly different types of FACTS controllers. [07]

OR

Q - 6 (a) What is arcing ground? On which system does it occur? [07]

(b) What type of lightning strokes can occur on transmission line? [07]

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